

Ok now could you summarize all the lit review we talked for me, including those that have similar pipeline with us, what they have done (those that ues CT/MTI as baseline) and what is the outcome from that research

Head-Model Morphing Literature Summary

Most of the literature we discussed clusters into **CT/MRI-driven personalization**, **template-to-subject morphing**, and the gap between those approaches and your **FLAME/MICA shell-only** idea. The strongest overall pattern is that imaging-based pipelines can preserve anatomy and mesh quality well enough for simulation, but they do that because they start with internal anatomy from CT or MRI (Giudice et al., 2020; Li et al., 2020; Henningsen et al., 2024).

Imaging-Based Pipelines

These studies are the closest established foundation for subject-specific head FE modeling. They start from segmented CT or MRI, register a baseline atlas or FE model, then propagate the displacement field into the mesh (Li et al., 2020; Henningsen et al., 2024; Zapf et al., 2025).

- Giudice et al. used **nonlinear MRI registration** to morph brain FE models for 44 subjects, and strain predictions differed by **less than 4%** from manually created voxel models (Giudice et al., 2020).
- Li and Kleiven built the **ADAPT personalization pipeline** from baseline and subject T1 MRI using Demons plus Dramms, and generated six subject-specific head models with satisfactory element quality and region-level DICE accuracy (Li et al., 2020).
- Henningsen et al. segmented **PMCT skull and intracranial layers**, morphed ADAPT to five forensic cases, and preserved subject-specific geometry and mesh quality in all cases (Henningsen et al., 2024).

Pipelines Similar To Yours

The papers most similar to your pipeline are not FLAME/MICA papers, but **surface- or template-driven morphing** papers that deform a validated baseline model to a target shape. They show the method is feasible, but also show why the head is the hardest region (Li et al., 2023; Zhang et al., 2017; Bucki et al., 2010).

Study Type	What They Did	Outcome
THUMS body morphing	Morphed THUMS and other HBMs to subject geometries by image registration	Personalized HBMs had comparable element quality, but naive THUMS head morphing produced an unusual head shape (Li et al., 2023)
THUMS avatar-driven morphing	Used RBF interpolation to adapt THUMS v6 across anthropometries	Chest and torso overlap was strong, and anatomical fidelity was broadly preserved with some localized distortion risk (Reyes et al., 2026)
Face-template FE morphing	Deformed a facial template with landmarks, dense fitting, and TPS	Generated high-quality patient-specific hexahedral meshes with good internal structure correspondence (Zhang et al., 2017)

Study Type	What They Did	Outcome
Atlas mesh registration	Registered a generic FE mesh to patient skin and bone surfaces	Succeeded in 60 cases with submillimetric surface accuracy, including 50 head CT cases (Bucki et al., 2010)

FIGURE 1 Pipelines most similar to shell-driven THUMS personalization

What CT/MRI Baselines Achieve

Across the CT/MRI-based papers, the main benefit is not just fitting the outside surface. The imaging provides **internal boundaries, ventricles, CSF spaces, white matter tracts, and tissue layers**, which lets the morphed model stay anatomically meaningful inside (Li et al., 2020; Huynh et al., 2025; Bahreinizad et al., 2024).

- Lifespan and pathology pipelines handled **newborn, elderly, and hydrocephalus brains** by adding feature-based and multimodality registration steps (Li, 2021).
- SynthMorph-based atlas morphing produced **300+ labeled structures** in about **20 minutes** while maintaining high Dice, low Hausdorff distance, and high mesh quality (Huynh et al., 2025).
- Earlier atlas-space meshing achieved **0.07 mm** average surface error and **99.5%** of elements with scaled Jacobian above 0.6, showing that registration-based meshing can be both accurate and fast (Ji et al., 2011).

Main Outcome For Your Project

The literature supports **morphing a validated baseline mesh to a subject shape**, but it does not yet show a paper that morphs **THUMS head directly to a FLAME/MICA outer shell** and then recovers a truly individualized internal head anatomy. The closest THUMS paper actually found that direct head morphing could fail because of topology differences, and solved that case by shielding the head rather than personalizing it (Li et al., 2023).

What the broader literature does support is a publishable **surface-constrained personalization** strategy: use the shell to drive outer-shape morphing, propagate the field inward smoothly, and treat the inside as **atlas-inherited anatomy** unless additional CT/MRI or statistical priors are added (Li, 2021; Zapf et al., 2025). That framing is also consistent with adjacent modeling work showing that realistic anatomy improves predictions, while surface conversion or coarse representations can lose important interface detail (Huang et al., 2017; Zhou et al., 2025; Tizzard et al., 2005).

So the short summary is: **CT/MRI-based pipelines are proven, template morphing is proven**, but **THUMS-to-FLAME shell-only individualized head modeling** still looks like an open research problem rather than an established method.

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